

Practice Quiz 02C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Finding a counterexample to an argument.

1. A student makes the argument

$$P \rightarrow Q, \quad \neg P, \quad \text{therefore } \neg Q.$$

Show that the argument is invalid by giving one assignment of truth values to P, Q . Evaluate both premises and the conclusion for your assignment, and explain why these values refute validity.

Solution Take $P = F$ and $Q = T$. Then $P \rightarrow Q$ is true, since its antecedent is false, and $\neg P$ is also true. The conclusion $\neg Q$ is false. Thus the argument can have both premises true and conclusion false, which is exactly a counterexample to validity. ■